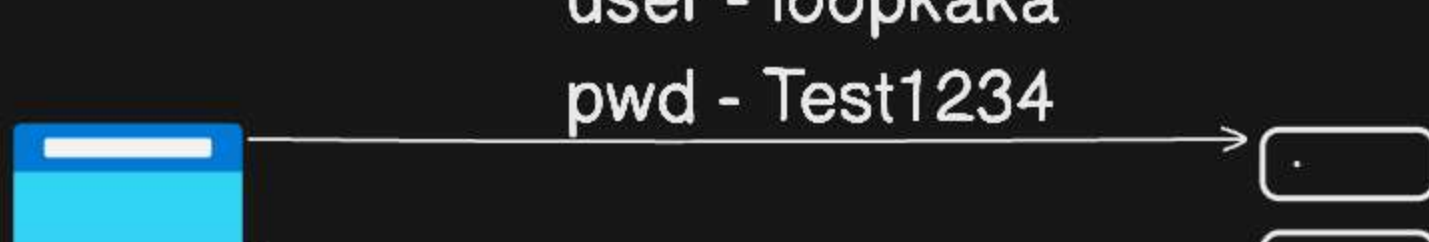


1. HTTP
2. MITM attack
3. HTTPS
4. SSL/TLS
5. Certificate
6. HSTS
7. Practical

HTTP (HyperText Transfer Protocol)

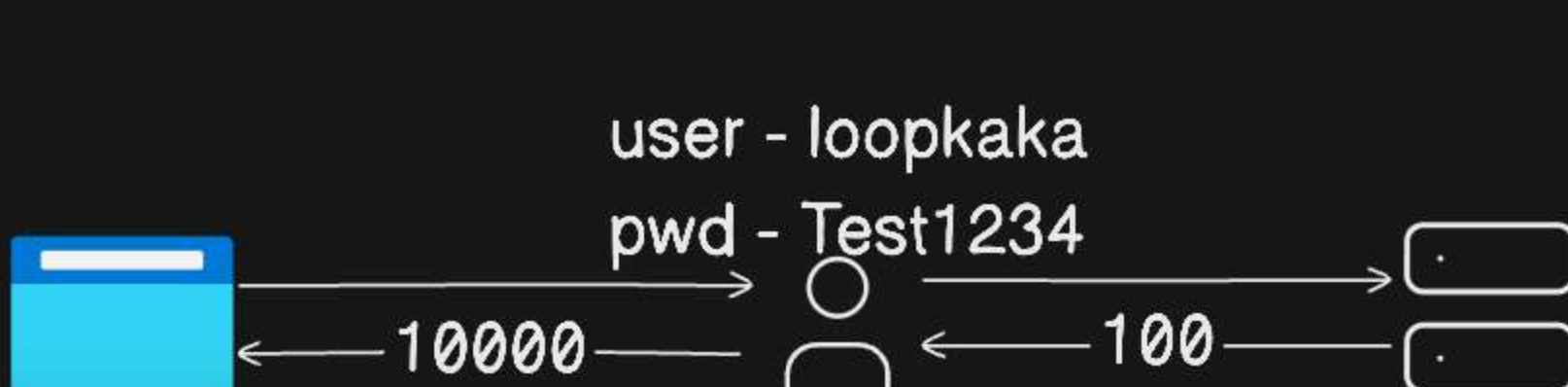
HTTP (HyperText Transfer Protocol) is a protocol used for communication between a browser and a server.



Problem in HTTP:

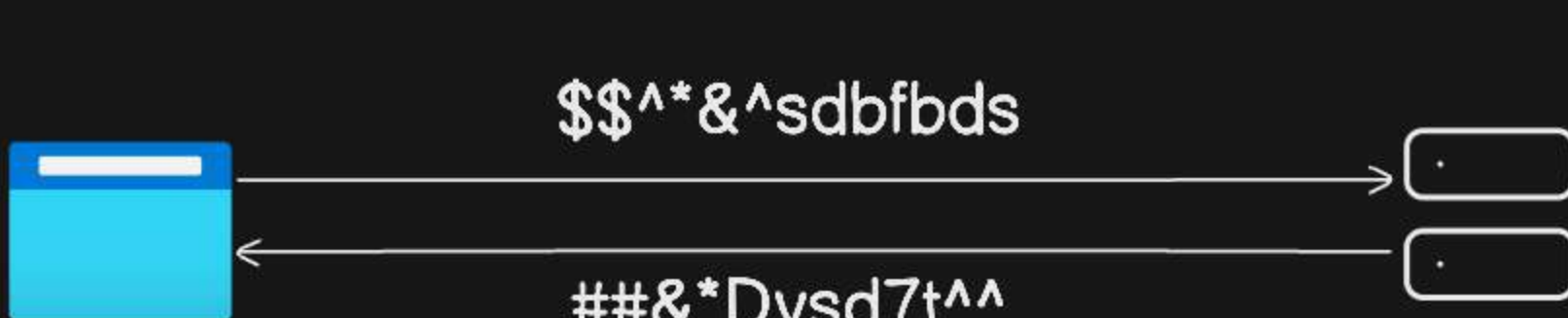
1. Evryting travles as plain text
2. Everyone can see your data
3. Hacker can do anything with your data

MITM Attack - (Man in the middlle attack)



HTTPS (HyperText Transfer Protocol Secure)

HTTPS = HTTP + Security



Security = SSL / TLS

SSL - Server Security Layer (outdated)

HTTPS is not SSL

TLS - Transport Layer Security

TLS replaces SSL

Benifits:

1. Encrypted - No one can see the data
2. Integrity - No one can change the data
3. Authentication - interact with correct server

Certificate:

Digital ID card for website

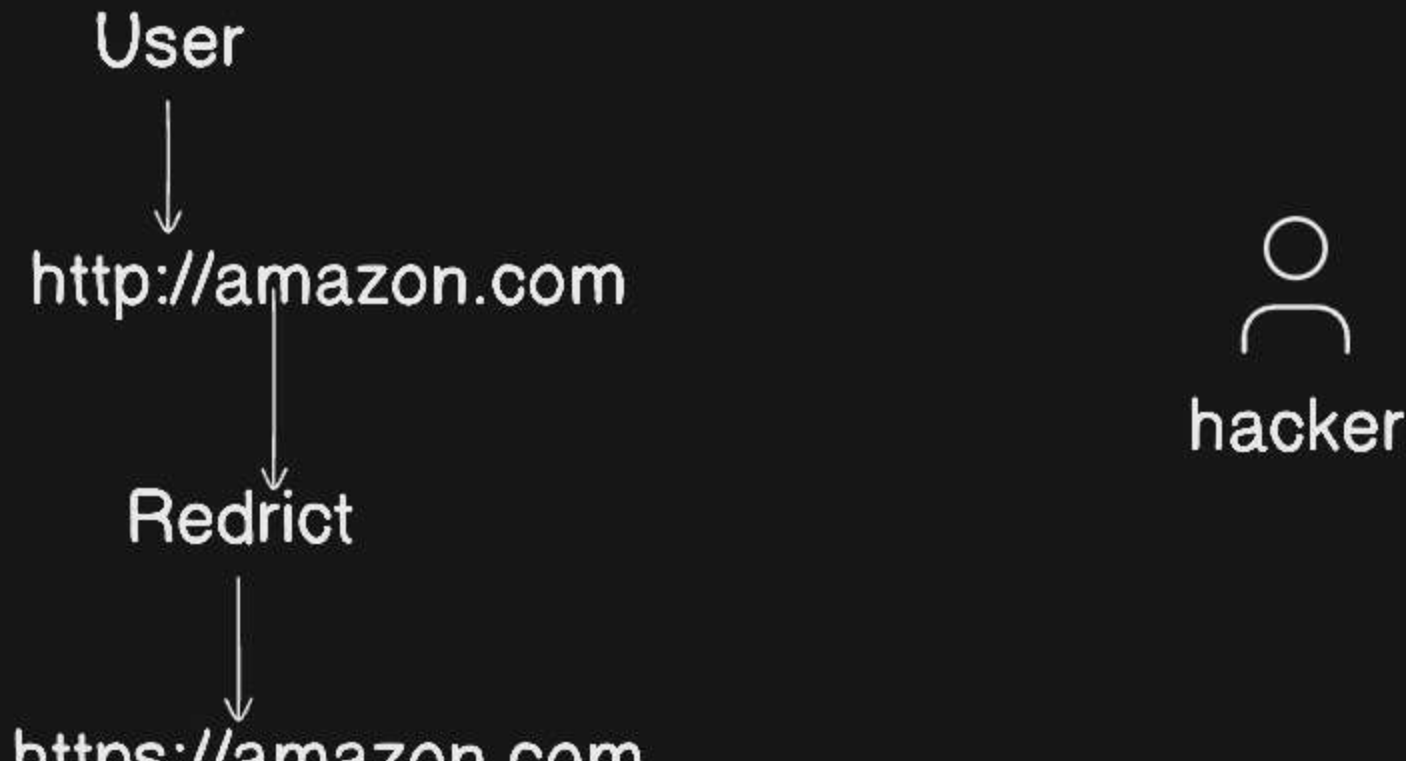
- website
- owner
- public key etc.

Certificate Authority:

- Lets Encrypt
- DigiCart
- GlobalSign

HSTS (HTTP Strict Tranport Security)

Without HSTS



Server sends header

strict-transport-security: max-age=31536000

You should always use HTTPS
for 1 year

Final Flow

